# THE FIVE COMPARISON PROPERTIES HOLD FOR THE PH-TOPOLOGY OVER THE SMALL ETALE SITE WITH PROPER-MORPHISM COEFFICIENTS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the proof is written out in full: the weak Serre verification with the exactness argument for the extension condition, and the five comparison properties with the reason and reference for each.

*Editorial note.* The label lemma-compare-ph-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source etale-cohomology.tex –label lemma-compare-ph-etale –start 22321 –end 22386 –local-id 3549 returned RESULT=OK (L3549 free, occupancy FREE). Candidate taken from lane-s4/halfB-priority.json.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- Cohomological descent for the etale comparison. Same chapter, the lemma stating that K -> Ra_{X,*}a_X^{-1}K is an isomorphism for torsion coefficients, giving exactness and fullness of pi_X^{-1}. Supplies the weak Serre verification and the identification of an extension with a pullback.
- The description of the pullback in the ph-setting. Same chapter, the lemma identifying pi_X^{-1} of a torsion sheaf with the rule given by the sections of its small pullback, in the ph-topology. Supplies property (item-A-sheaf).
- Base change, torsion and the ph-covering property of proper morphisms. Morphisms, the permanence of properness under base change; Same chapter, the comparison of the small and big higher direct images and the torsion of direct images; Topologies, the lemma that a surjective proper morphism is a ph-covering; More on Morphisms, the domination of an fppf cover by a finite one. Supplies properties (item-base-change-P), (item-A-and-P) and (item-refine-tau-by-P).

*Source reference.* Original lemma 0754: sources/stacks-chow/tags.txt, source lines 9118–9118. Prerequisite pointer invoked at proof line(s) 22344; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original reference 0DDI: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21308–21308. Prerequisite pointer invoked at proof line(s) 22346; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0DDJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21323–21323. Prerequisite pointer invoked at proof line(s) 22374; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0DDW: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 22175–22175. Prerequisite pointer invoked at proof line(s) 22370; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0DDD: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 13924–13924. Prerequisite pointer invoked at proof line(s) 22377; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0DET: sources/stacks-chow/tags.txt, source lines 16813–16813. Prerequisite pointer invoked at proof line(s) 22382; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 01W4: sources/stacks-chow/tags.txt, source lines 2469–2469. Prerequisite pointer invoked at proof line(s) 22361; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0EZ7: sources/stacks-chow/tags.txt, source lines 18717–18717. Prerequisite pointer invoked at proof line(s) 22335; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0EZ6: sources/stacks-chow/tags.txt, source lines 18716–18716. Prerequisite pointer invoked at proof line(s) 22334; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0EZ4: sources/stacks-chow/tags.txt, source lines 18714–18714. Prerequisite pointer invoked at proof line(s) 22332; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0EZ8: sources/stacks-chow/tags.txt, source lines 18718–18718. Prerequisite pointer invoked at proof line(s) 22336; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0EZ5: sources/stacks-chow/tags.txt, source lines 18715–18715. Prerequisite pointer invoked at proof line(s) 22333; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0EZ3: sources/stacks-chow/tags.txt, source lines 18713–18713. Prerequisite pointer invoked at proof line(s) 22338; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 021I: sources/stacks-chow/tags.txt, source lines 2671–2671. Prerequisite pointer invoked at proof line(s) 22366; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0DES: sources/stacks-chow/tags.txt, source lines 16812–16812. Prerequisite pointer invoked at proof line(s) 22385; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Consider the comparison morphism $\epsilon : (Sch/S)_{ph} \to (Sch/S)_{étale}$. Let $\mathcal{P}$ denote the class of proper morphisms of schemes. For $X$ in $(Sch/S)_{étale}$ denote $\mathcal{A}'_X \subset Ab((Sch/X)_{étale})$ the full subcategory consisting of sheaves of the form $\pi_X^{-1}\mathcal{F}$ where $\mathcal{F}$ is a torsion abelian sheaf on $X_{étale}$ Then Cohomology on Sites, Properties (0EZ4), (0EZ5), (0EZ6), (0EZ7), and (0EZ8) of Cohomology on Sites, Situation 0EZ3 hold.*

*Proof.* We first show that $\mathcal{A}'_X \subset Ab((Sch/X)_{étale})$ is a weak Serre subcategory by checking conditions (1), (2), (3), and (4) of Homology, Lemma 0754. Parts (1), (2), (3) are immediate as $\pi_X^{-1}$ is exact and fully faithful for example by Lemma 0DDI. If $0 \to \pi_X^{-1}\mathcal{F} \to \mathcal{G} \to \pi_X^{-1}\mathcal{F}' \to 0$ is a short exact sequence in $Ab((Sch/X)_{étale})$ then $0 \to \mathcal{F} \to \pi_{X,*}\mathcal{G} \to \mathcal{F}' \to 0$ is exact by Lemma 0DDI. In particular we see that $\pi_{X,*}\mathcal{G}$ is an abelian torsion sheaf on $X_{étale}$. Hence $\mathcal{G} = \pi_X^{-1}\pi_{X,*}\mathcal{G}$ is in $\mathcal{A}'_X$ which checks the final condition.

Cohomology on Sites, Property (0EZ4) holds by the existence of fibre products of schemes and the fact that the base change of a proper morphism of schemes is a proper morphism of schemes, see Morphisms, Lemma 01W4.

Cohomology on Sites, Property (0EZ5) follows from the commutative diagram (3) in Topologies, Lemma 021I.

Cohomology on Sites, Property (0EZ6) is Lemma 0DDW.

Cohomology on Sites, Property (0EZ7) holds by Lemma 0DDJ part (2) and the fact that $R^i f_{small}\mathcal{F}$ is torsion if $\mathcal{F}$ is an abelian torsion sheaf on $X_{étale}$, see Lemma 0DDD.

Cohomology on Sites, Property (0EZ8) follows from More on Morphisms, Lemma 0DET combined with the fact that a finite morphism is proper and a surjective proper morphism defines a ph covering, see Topologies, Lemma 0DES. □